



COLLEGE OF ENGINEERING



Fuels



Carbonization of Coal

- The process of converting coal into coke is called carbonization.
- Heating the coal in absence of air, to a sufficiently high temperature, so that coal undergo decomposition to yield a residue.
- The residue is richer in carbon content than the original fuel.
- This residue is a lustrous, dense, strong, porous and coherent mass, known as coke.

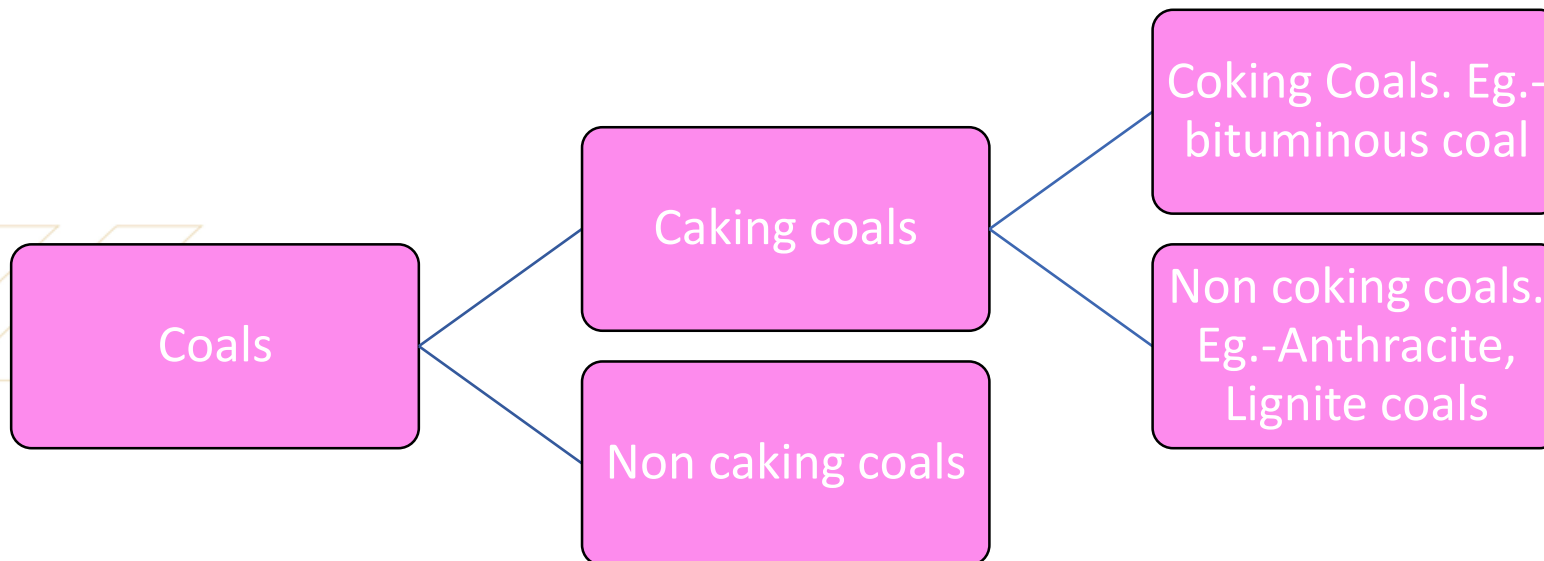
Caking and Coking of Coals

Caking coals:

- The coals that have tendency to soften and swell at higher temperature to form a solid coherent mass with porous structure are called caking coals. The residue thus formed is called coke.

Coking coals:

- If the coke is hard, porous and strong than the coal from which it is formed is called coking coal. This property is found only in bituminous type of coal.



Process of Carbonization

1. First, moisture and occluded gases are driven off.
2. $\sim 260^{\circ}\text{C}$ - 270°C , Carbon, water & H_2S , some low molecular alkenes and alkanes are evolved.
3. $\sim 350^{\circ}\text{C}$ decomposition of coal is accompanied by evolution of gases and elimination of tarry vapors.
4. Around 400°C , caking coals become soft & plastic.
5. At about 700°C , hydrogen is evolved.
6. At about 800°C , main gaseous products are evolved.
7. Gases evolved form plastic mass, expand it to give foam like appearance
8. At further high temperature, this foam like mass solidifies to form solid mass with porous structure called coke.

Types of Carbonization

The properties of the coke produced mainly depend on the type of coal used, temperature of carbonization and the rate of carbonization. Depending upon the temperature upto which coal is heated in absence of air, the carbonization is of two types:

1. Low temperature carbonization (LTC)
2. High temperature carbonization (HTC)

Types of Carbonization

Characteristics	Low temperature carbonization	High temperature carbonization
Carbonization temperature	500-700 ⁰ C	1000-1200 ⁰ C
Yield	75-80%	65-75%
Volatile matter	5-15%	1-3%
Mechanical strength	Not mechanically strong	Mechanically strong
Hardness	Soft coke	Hard coke
Uses	For domestic purposes	For metallurgical purposes

Types of Carbonization

Characteristics	Low temperature carbonization	High temperature carbonization
Yield of the gas	Lower (About 130-150 m ³ /tonne)	Higher (About 300-390m ³ /tonne)
Calorific value of the gas	Higher (6500-9500 kcal/m ³)	Lower (5400-6000 kcal/m ³)
% of straight chain hydrocarbons in the byproducts	Higher	Lower
% of aromatics in the byproducts	Lower	Higher
Smoke produced on burning	Smokeless	Smoky

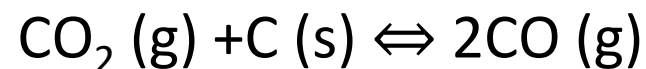
Metallurgical Coke

Metallurgical coke: The properties of coke depend on porosity, reactivity and the amount of volatile matter retained by coke during carbonization.

- Coke is mainly used as a heat source and reducing agent in metallurgy. A good coke in metallurgical process should possess the following characteristics:
 - **Purity:** The metallurgical coke should contain lower percentage of moisture, ash, sulphur and phosphorous.
 - **Porosity:** The coke should be porous so as to provide contact between carbon and oxygen which ensures efficient and complete combustion of the fuel.
 - **Strength:** The coke used in metallurgical process should have high strength so as to withstand the weight of the ore, flux etc. in the furnace.

Metallurgical Coke

- **Size:** Metallurgical coke should be of medium size.
- **Combustibility:** Coke should burn easily. The combustibility of coke depends on the nature of the coal, carbonization temperature and reaction temperature.
- **Calorific value:** It should be high.
- **Cost:** Coke should be cheap and easily available near the site of the metallurgical plant.
- **Reactivity:** Reactivity of coke is its ability to react with CO₂ , steam, air and oxygen. The reactivity should not be too high. The reactivity toward CO₂ represent the reduction of CO₂ according to equation



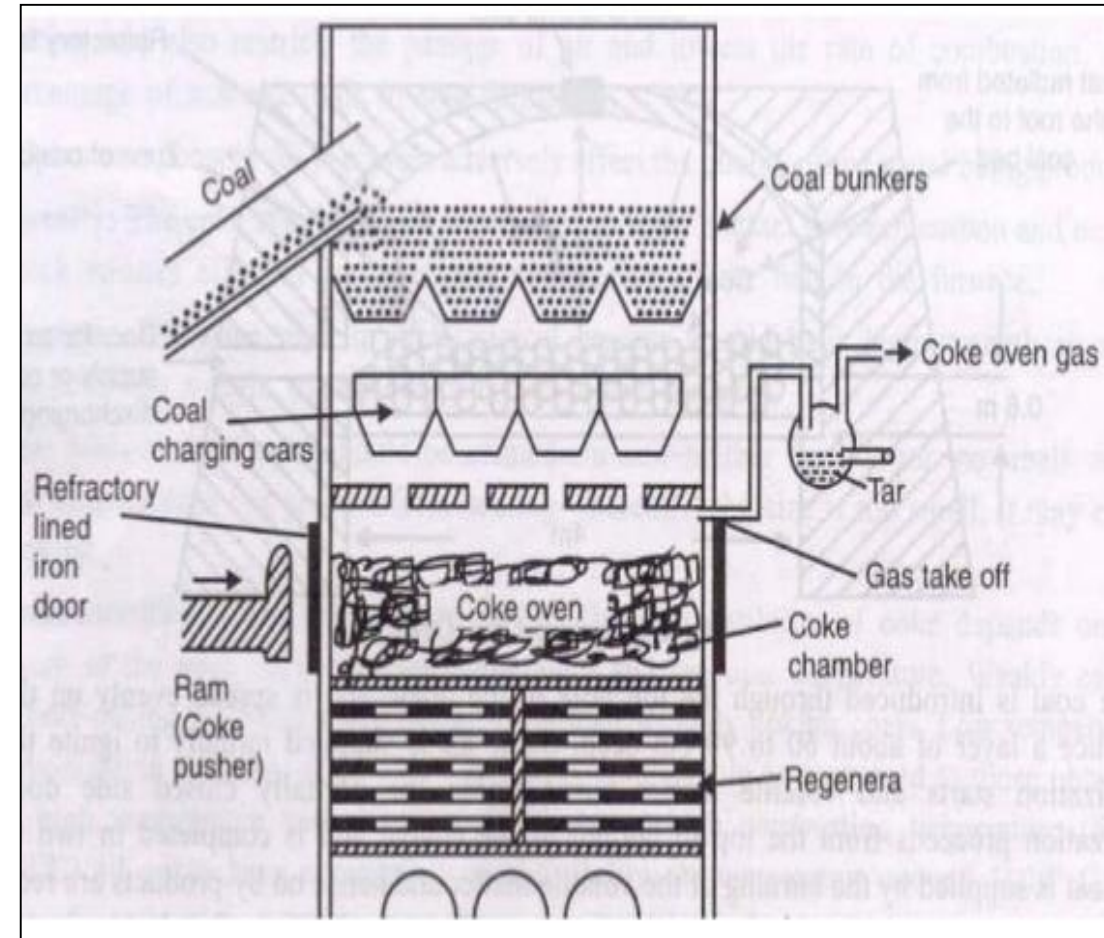
Manufacture - Metallurgical Coke

Otto-Hoffmann's oven or By-product Oven

These chamber ovens work on regenerative principle of heat economy i.e. utilization of waste fuel gases for heating of the checker work of bricks. All the valuable products are recovered from the outgoing flue gases.

Construction: Otto-Hoffmann Oven

- It consists of no. of narrow rectangular chambers made of silica bricks.
- The usual dimensions are 12 to 14 m length, 4 to 5 m height and 0.5 m width. The chambers are tightly closed so that no air can enter.
- At the top of each chamber there are three holes for charging the coal.
- Each chamber is further provided with a gas take off and refractory lined cast iron door at each end for discharging the coke.
- The oven form a sort of battery of 10-100 ovens.
- One single oven is capable of holding 16-24 tonnes of coal.



Working : Otto-Hoffmann Oven

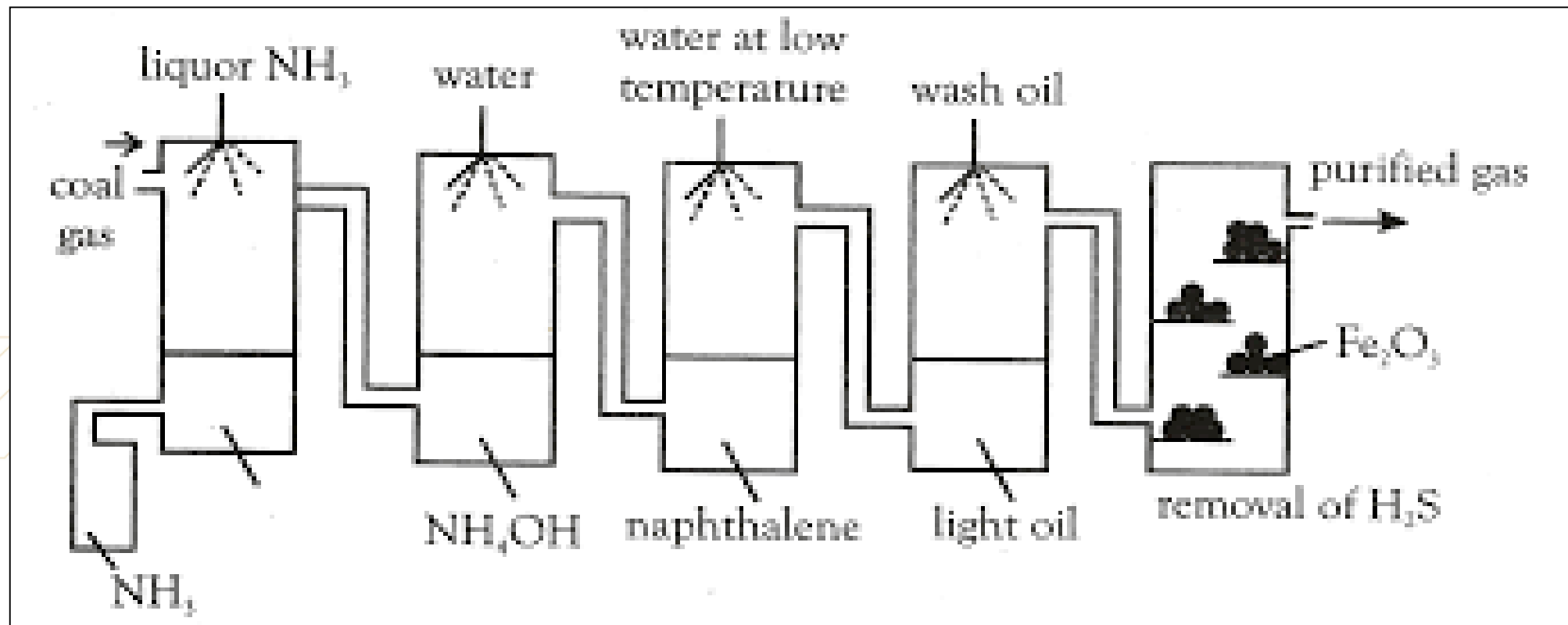
- Coal is charged into the chamber and the chambers are closed tightly at both the ends.
- The coke ovens are heated to 1200°C by burning gaseous fuels.
- The process of carbonization takes place layer by layer in the coal charge starting from the two side walls of the oven and moving towards the center. As the coal adjacent to the oven walls gets heated, a plastic zone is formed which moves away from the walls towards the central zone.
- As the coal is converted into coke, there is decrease in volume. This is because of the removal of volatile matter in the form of tar and gas at about 500°C.
- At further high temperature, the plastic mass solidifies into hard and porous mass called coke.
- Regenerative principle is employed to achieve as economical heating as possible.

Working : Otto-Hoffmann Oven

- Regenerators are built underneath the ovens.
- The flue gases pass their heat to the checker brick work of regenerators until the temperature rises to 1000°C.
- Regenerators work on the principle of alternate heating and cooling cycles.
- This is achieved by periodically changing the direction of flow of gases through the vertical flues every 30 min or so.
- Carbonization of a charge of coal takes about 11-18 hours. After the process is complete, red hot coke is pushed outside by means of a ram which is electrically driven. The coke falls into a quenching car.
- The yield of the coke is 75 % of coal.

Recovery of by-products

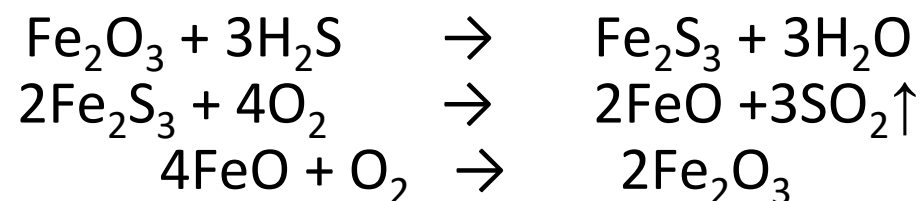
- The gases and vapours evolved on carbonization in coke ovens are not allowed to mix with the combustion and are collected separately.
- The coke oven gas is treated separately for the recovery of the valuable byproducts.



Recovery of by-products

1. **Recovery of Tar:** The gas from the coke ovens is passed through a tower in which liquor ammonia is sprayed. Tar and dust get collected in a tank. The tank is provided with a heating coils to recover back ammonia.
2. **Recovery of ammonia:** The gases are then passed through a tower where water is sprayed to recover ammonia. The ammonia can also be recovered by dissolving it in H_2SO_4 to form $(\text{NH}_4)_2\text{SO}_4$, which is then used as a fertilizer.
3. **Recovery of Naphthalene:** The gases are passed through a cooling tower, where water at a low temperature is sprayed. The gas is scrubbed with water until its temp. reduces. The naphthalene gets condensed here which is collected along with water and recovered.

- **Recovery of Benzol:** The gases are then introduced into a light oil or benzol scrubber, where benzene along with its homologue is removed and is collected at the bottom.
- **Recovery of H₂S and other S compounds:** H₂S and other S compounds are removed from the coke oven gas after the light oil has been separated out. For the removal, the gas is passed through a chamber, packed with moist Fe₂O₃, which reacts with H₂S to give Fe₂S₃.



- The SO₂ obtained can be used for the manufacturing of sulphuric acid, which can be used to absorb NH₃ from the coal gas.

THANK YOU

